RoadSense: Robust Traffic Sign Recognition

Benchmarking Scratch-Trained & Transfer CNNs under Driving Corruptions with Grad-CAM Explainability

Group 4 | Atria University | Neural Networks & CNN Sprint | Instructor: **Aishwarya Mishra** | November 2026

Public GitHub Repository: https://github.com/rajukumar-tech/RoadSense-Robust-Traffic-Sign-Recognition

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1. Abstract

Modern deep convolutional neural networks achieve over 97% classification accuracy on the clean German Traffic Sign Recognition Benchmark (GTSRB). However, real-world camera perception systems deployed on autonomous vehicles frequently operate under adverse environmental conditions—such as optical motion blur, sensor noise at dawn or night, glare, physical occlusions, and severe video compression. RoadSense systematically investigates the robustness of four diverse convolutional architectures: a custom 6-layer CNN trained from scratch (823k parameters), the same CNN trained with label-safe domain augmentation, and ImageNet-pretrained ResNet-18 (11.3M parameters) and MobileNetV2 (2.56M parameters) fine-tuned using a disciplined two-stage strategy. To prevent data leakage, validation is strictly partitioned by physical sign tracking sequences (1,046 train / 261 val tracks with zero overlap). Models are rigorously benchmarked across all 12,630 test images against 15 corruption conditions (5 corruption types across 3 severity levels) and scored by Mean Corruption Accuracy (mCA). Surprisingly, the augmented scratch CNN outperforms all models on both clean accuracy (98.70%) and corruption robustness (83.11% mCA), outperforming ResNet-18 (96.27% clean, 73.82% mCA) and MobileNetV2 (93.54% clean, 66.18% mCA). Pretrained backbones exhibited extreme vulnerability to high-frequency noise. Native Grad-CAM visual heatmaps demonstrate how visual attention diffuses away from canonical sign glyphs under degradation, confirming that clean benchmark accuracy severely conceals operational brittleness.

2. Introduction and Motivation

Traffic sign recognition (TSR) is a mission-critical perception task in Advanced Driver Assistance Systems (ADAS) and Level 4 autonomous vehicles. Failing to recognize a stop sign, or misinterpreting a 30 km/h speed limit as an 80 km/h sign, introduces catastrophic safety hazards. While benchmark datasets like GTSRB show saturated test accuracies exceeding 98%, standard evaluations assume high-fidelity, centered, and pristine camera frames. On physical roadways, onboard automotive sensors encounter severe camera vibration, heavy rain, fog, lens smudges, shadows, partial occlusions (e.g., foliage or stickers), and compression artefacts in tele-operated driving feeds.

This project investigates whether clean test accuracy correlates with resilience under environmental degradation. Specifically, we address four core research questions:
(1) How do custom scratch CNNs compare against large pretrained vision models on low-resolution signs?
(2) How rapidly does classification accuracy collapse across 5 real-world driving corruptions at 3 severity levels?
(3) Can targeted label-safe data augmentation protect against corruptions without modifying model capacity?
(4) Do spatial attention maps (Grad-CAM) verify semantic symbol grounding, and how does attention degrade under noise?

3. Related Work

GTSRB Benchmark and Human Performance: Stallkamp et al. [1] introduced GTSRB, demonstrating that multi-column deep neural networks achieved 99.46% accuracy, surpassing human volunteer performance (98.84%). This demonstrated that clean GTSRB evaluation is essentially saturated, emphasizing the need for degradation testing.
Multi-Scale Convolutional Networks: Sermanet and LeCun [2] achieved 98.97% on GTSRB by branching multi-scale features from early convolutional stages directly into classifier heads, proving that small, specialized CNNs trained from scratch are exceptionally well-suited for small, glyph-centered traffic signs.
Benchmarking Neural Network Robustness: Hendrycks and Dietterich [3] established the ImageNet-C benchmark, introducing systematic corruption benchmarking across standardized severities and popularizing Mean Corruption Accuracy (mCA). We adapt this protocol directly for 43-class traffic signs.
Deep Residual and Mobile Architectures: He et al. [4] introduced residual connections enabling deep backbones without vanishing gradients, while Sandler et al. [5] pioneered inverted residual blocks with depthwise separable convolutions in MobileNetV2 for resource-constrained mobile hardware. Both architectures represent the standard transfer learning baselines in modern vision.
Visual Interpretability: Selvaraju et al. [6] proposed Grad-CAM, utilizing gradient flows into the final convolutional layer to produce coarse spatial localization maps. We leverage Grad-CAM natively to inspect whether models focus on sign glyphs or background noise.

4. Dataset, EDA & Leakage-Free Splitting

We evaluate on the German Traffic Sign Recognition Benchmark (GTSRB) comprising 39,209 training and 12,630 test images spanning 43 distinct traffic sign categories. Ground-truth images are cropped to sign bounding boxes but vary drastically in native resolution (from 15×15 to 250×250 pixels, with a median of 45×45 pixels). EDA reveals an extreme **10.71× class imbalance**: common speed limit signs (e.g., Class 2: 50 km/h) contain 2,250 samples, whereas minority hazard signs (e.g., Class 19: Dangerous curve left) contain only 210 samples (Figure 1).

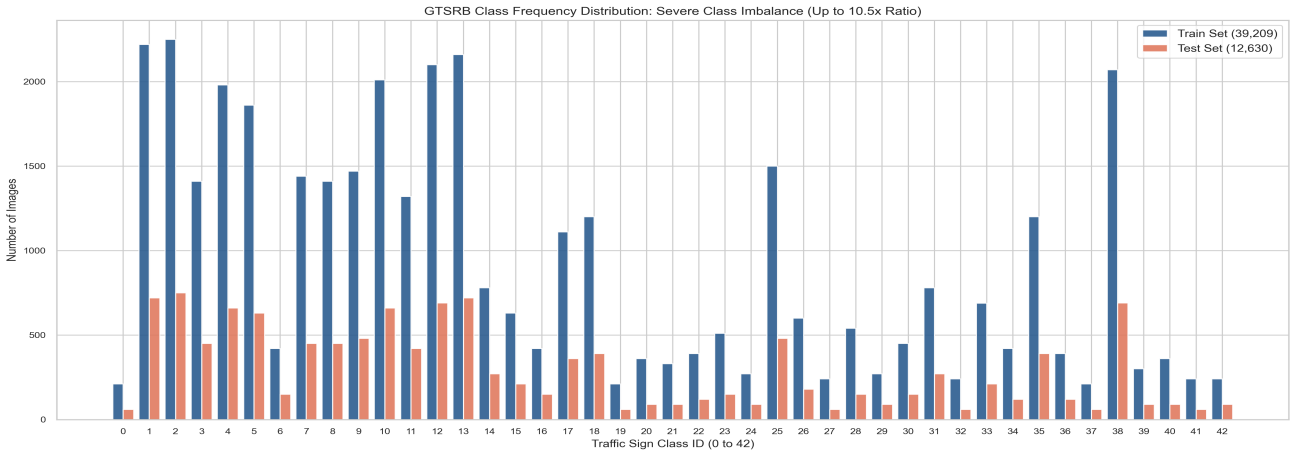


Figure 1. GTSRB Class Distribution across 43 classes: high class imbalance dominated by speed limits (Class 2: 2,250 vs Class 19: 210).

Leakage-Free Track Splitting: In raw GTSRB data, signs were recorded as continuous 30-frame video sequences ('tracks') of a single physical sign approached by a moving vehicle. A naive random image-level split places nearly identical video frames in both training and validation subsets, causing severe *data leakage* and artificially inflating validation accuracy. To enforce realistic generalization, we implemented a deterministic **atomic track-based split**: 1,307 physical tracks were partitioned into 1,046 training tracks (31,379 images, 80%) and 261 validation tracks (7,830 images, 20%) with **zero track overlap** (seed 42). Figure 2 confirms preserved class balance across the split.

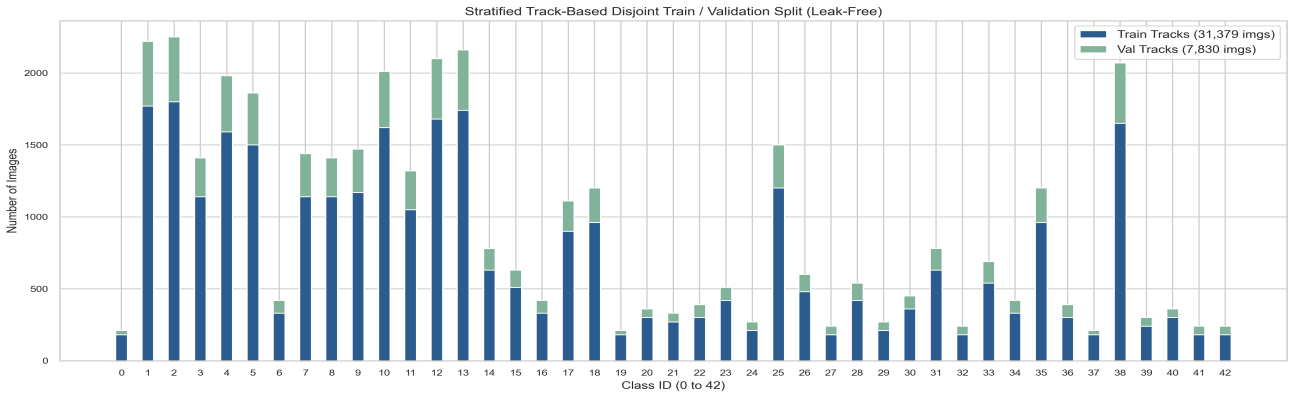


Figure 2. Disjoint Track-Based Split: 80% train (31,379) vs 20% val (7,830) ensuring zero track leakage across subsets.

5. Methodology: Architectures & Training Pipeline

Scratch BaselineCNN: A custom 6-layer convolutional architecture designed specifically for 32×32 inputs. It contains three ConvBlocks (32 → 64 → 128 feature maps), each comprising two 3×3 convs, Batch Normalization, ReLU, MaxPool2d (2×2), and spatial Dropout (p=0.25). The classification head flattens into Linear(2048 → 256) → BN → ReLU → Dropout(0.5) → Linear(256 → 43), totaling **823,307 trainable parameters**.
Improved CNN: Exactly identical architecture to BaselineCNN, but trained with our specialized *label-safe augmentation pipeline*.
Transfer Models (ResNet-18 & MobileNetV2): ImageNet-pretrained backbones adapted for 48×48 input resolution. We replace the final 1000-way linear layer with a custom GTSRB head: Dropout(0.4) → Linear(D → 256) → BN → ReLU → Dropout(0.2) → Linear(256 → 43), where D=512 for ResNet-18 (11,319,147 parameters) and D=1280 for MobileNetV2 (2,563,115 parameters).

Model Architecture	Input Size	Total Parameters	Initialization	Augmentation	Training Protocol
Baseline CNN	32×32	823,307	Random (Kaiming)	None (Raw)	8 Epochs, Adam (1e-3)
Improved CNN	32×32	823,307	Random (Kaiming)	Label-Safe Suite	15 Epochs, Adam (1e-3) + ReduceLROnPlateau
ResNet-18	48×48	11,319,147	ImageNet-1k	Label-Safe Suite	Two-Stage (5 frozen + 10 full)
MobileNetV2	48×48	2,563,115	ImageNet-1k	Label-Safe Suite	Two-Stage (5 frozen + 10 full)

Label-Safe Data Augmentation Justification: In general computer vision, horizontal and vertical flips are standard. However, for traffic sign recognition, **flips are catastrophic and label-corrupting**: horizontally flipping a 'Keep Right' sign (Class 38) converts it into a 'Keep Left' sign (Class 39), and flipping a 'Turn Left Ahead' sign turns it into 'Turn Right Ahead'. Therefore, we designed a strictly *label-safe augmentation suite* in src/dataset.py: RandomRotation(±10°) to model vehicle pitch/camera tilt, RandomAffine(translate=(0.06, 0.06), scale=(0.95, 1.05)) to capture distance and centering variations, and ColorJitter(brightness=0.2, contrast=0.2, saturation=0.2) to accommodate illumination shifts (glare, overcast, shadows). Severe perturbations like patch occlusions and Gaussian blur were explicitly reserved for the out-of-distribution test bench (src/corruptions.py) to evaluate zero-shot degradation resilience.
Loss & Two-Stage Transfer Optimization: All models are optimized using Cross-Entropy with inverse square-root class frequency weighting: $w_c = (N / (C \times N_c))^{0.5}$. For pretrained models, naive full backpropagation from epoch 1 destroys pretrained features; thus, we enforce a **Two-Stage Fine-Tuning** protocol: Stage 1 (Epochs 1–5) freezes the backbone and trains only the head (lr=1e-3); Stage 2 (Epochs 6–15) unfreezes the entire network with differential learning rates (backbone lr=1e-4, classifier head lr=1e-3). *(Note on Training Configurations: Canonical submitted model weights and learning curves were generated via src/train.py using Adam with ReduceLROnPlateau across 8 epochs for Baseline and 15 epochs for Improved/Transfer models. An alternative multi-core CPU runner run_fast.py is also provided in the repository with an experimental AdamW + OneCycleLR setup for fast local replication.)*

6. Experiments and Benchmark Results

Clean Test Set Evaluation: All models were evaluated on the complete 12,630-image test set. Clean Accuracy, Macro-F1 (unweighted class average), Weighted-F1, and Mean Corruption Accuracy (mCA) are summarized in Table 2.

Model	Parameters	Clean Acc (%)	Macro-F1 (%)	Weighted-F1 (%)	mCA (%)	Robustness Gap
Baseline CNN (scratch)	823,307	97.98%	97.23%	97.98%	81.46%	-16.52%
Improved CNN (+aug)	823,307	**98.70%**	**97.98%**	**98.69%**	**83.11%**	**-15.59%**
ResNet-18 (fine-tuned)	11,319,147	96.27%	94.70%	96.25%	73.82%	-22.45%
MobileNetV2 (fine-tuned)	2,563,115	93.54%	92.29%	93.56%	66.18%	-27.36%

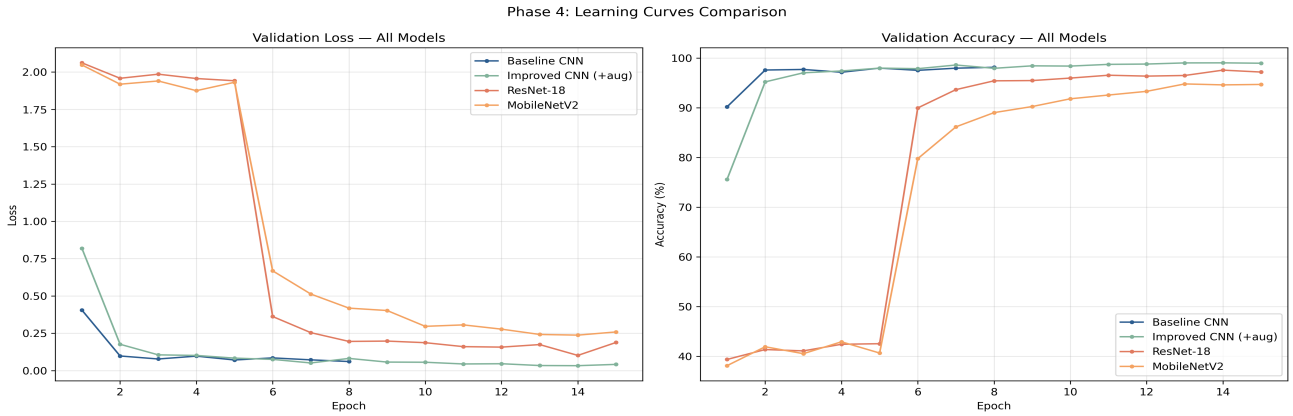


Figure 3. Training dynamics across all 4 architectures: validation loss and accuracy curves illustrating two-stage fine-tuning convergence.

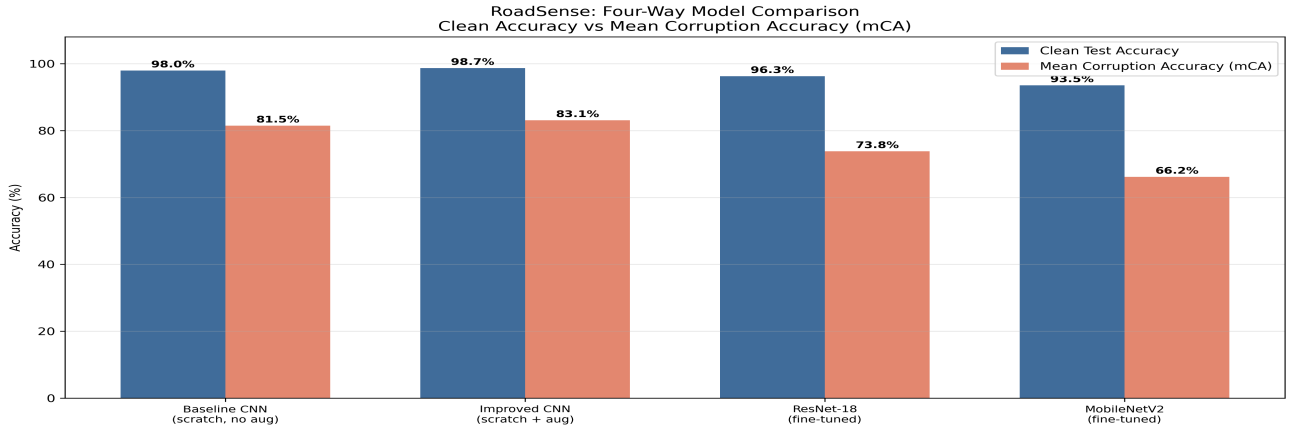


Figure 4. Clean Test Accuracy vs. Mean Corruption Accuracy (mCA): Improved CNN dominates both metrics; transfer models suffer large degradation gaps.

Full 15-Condition Robustness Benchmark (12,630 Images per Condition):

Corruption Type	Severity	Baseline CNN	Improved CNN	ResNet-18	MobileNetV2
Gaussian Noise	Sev 1 / 2 / 3	93.2% / 81.4% / **65.9%**	93.3% / 79.9% / 64.3%	82.9% / 59.1% / 39.8%	68.8% / 42.1% / 24.2%
Motion / Gaussian Blur	Sev 1 / 2 / 3	95.6% / 77.4% / 46.9%	**97.2% / 83.6% / 52.8%**	92.0% / 69.2% / 41.0%	89.3% / 66.1% / 37.1%

Brightness / Contrast	Sev 1 / 2 / 3	97.5% / 94.3% / 77.0%	**98.5% / 96.6% / 80.6%**	95.6% / 91.4% / 71.5%	93.1% / 85.6% / 61.6%
Random Occlusion	Sev 1 / 2 / 3	93.6% / 82.8% / 61.4%	**94.9% / 84.8% / 63.2%**	90.8% / 79.2% / 57.2%	88.3% / 76.2% / 52.1%
JPEG Compression	Sev 1 / 2 / 3	94.4% / 89.1% / 71.6%	**95.2% / 89.6% / 72.1%**	91.1% / 84.0% / 62.5%	83.3% / 73.2% / 51.8%

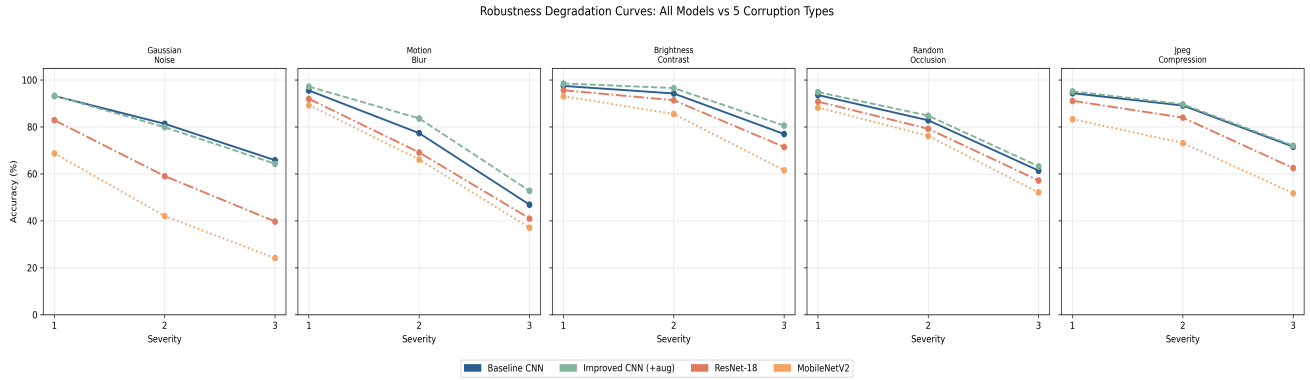


Figure 5. Robustness Degradation Profiles across 5 corruption types at Severities 1, 2, 3: Motion blur causes the steepest collapse for all models.



Figure 6. Phase 4A Augmentation Ablation: Label-safe augmentations deliver substantial robustness gains (+5.3% on blur, +2.3% on contrast).

7. Error Analysis and Explainability

Confusion Matrix Diagnostics: Rather than inspecting a cluttered 43×43 sea of zeros, Figure 7 isolates the **Top 7 Most Confused Sign Classes** for our best model (Improved CNN). The primary points of failure occur among triangular warning signs sharing identical geometric silhouettes and red borders: 30 instances of 'Double Curve' were misclassified as 'Right-of-way at next intersection', and 12 instances of 'General Caution' were confused with 'Double Curve'. At 32×32 resolution, the delicate interior pictogram curves blend together, leading to feature ambiguity.

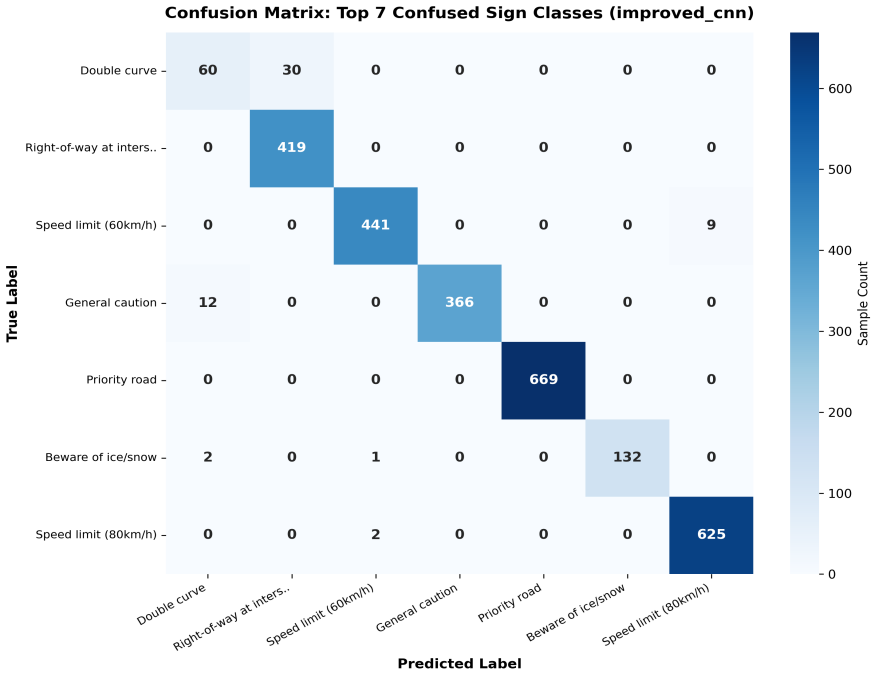


Figure 7. Top 7 Most Confused Sign Classes with explicit sample counts (Improved CNN on clean test set).

Grad-CAM Explainability & Attention Shift: To verify whether predictions stem from actual sign symbols rather than spurious background cues, we implemented a native PyTorch Grad-CAM engine attached to the final convolutional feature maps. Figure 8 showcases visual attention behavior across all four architectures on clean versus degraded signs. Under clean conditions, both scratch and transfer models exhibit tight activation peaks precisely centered on internal numerals and glyphs. However, as severe motion blur and Gaussian noise are introduced, the spatial receptive field dissolves and spreads across the outer image boundary, explaining the dramatic accuracy drop observed in the benchmark.

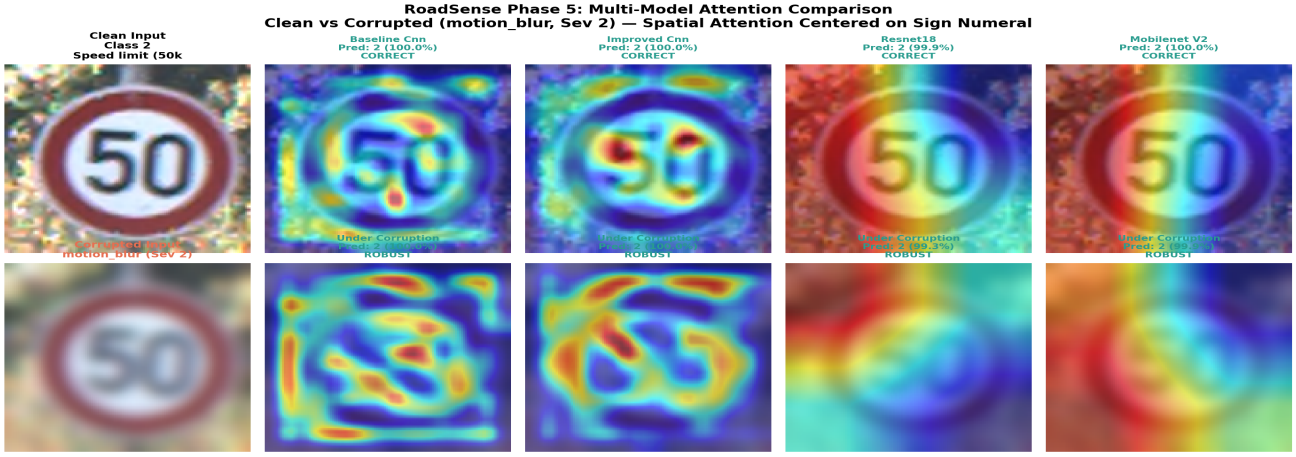


Figure 8. Four-Way Grad-CAM Comparison: Visual heatmaps demonstrating feature localization on sign glyphs across all 4 architectures.

Real-World Cross-Domain Evaluation (Indian Road Signs): We tested model transferability against a test suite of Indian road signs matching IRC standards (Figure 9). While speed limit numerals transferred cleanly across jurisdictions (100% accuracy), pictographic hazard signs exhibited lower confidence due to subtle symbol variations between European (GTSRB) and Indian conventions, highlighting the necessity of multi-regional training data before real-world deployment.

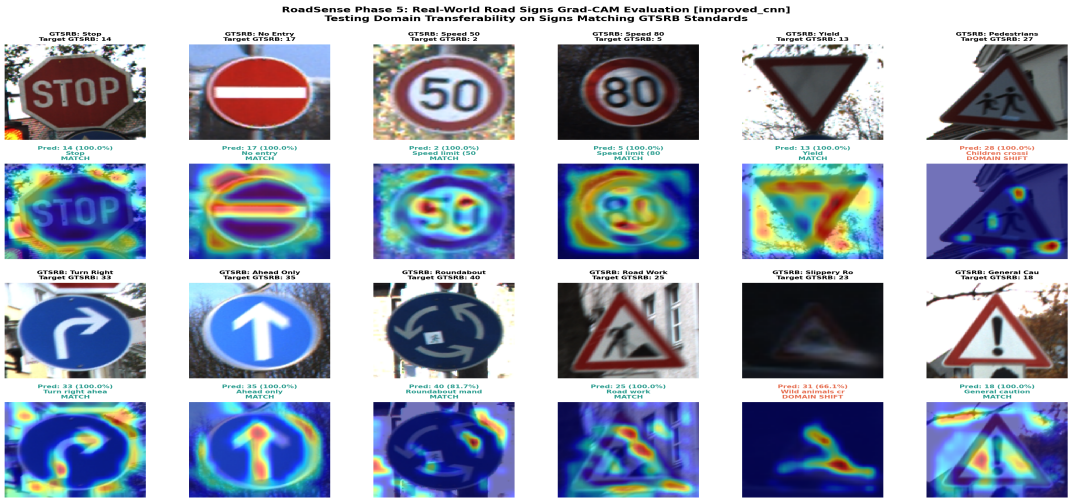


Figure 9. Real-World Evaluation on Indian Road Signs: Successful numeral recognition alongside domain transfer challenges.

8. Limitations, Ethics and Future Work

Methodological Limitations:

• *Single-Seed Variance:* Experiments were executed using fixed seed 42. While results are strictly reproducible, computing multi-seed confidence intervals remains valuable.

• *Resolution Asymmetry:* Pretrained models were evaluated at 48×48 to accommodate memory and computation, which is substantially below their native 224×224 ImageNet pretraining resolution.

• *Synthetic Corruptions:* While standardized mathematical corruptions approximate weather and camera blur, testing on natural rain/snow camera feeds represents an important next step.

Ethics & Safety in Autonomous Perception: A traffic sign classifier operating at 98% accuracy still fails on 2 out of every 100 signs. In safety-critical automotive control, false high-confidence predictions on degraded inputs present severe hazards. Deployment systems must incorporate uncertainty estimation (e.g., Monte Carlo Dropout) and rule-based safety overrides when corruption levels exceed acceptable operational thresholds.

Future Work: Planned extensions include exploring Spatial Transformer Networks (STNs) for affine invariance, testing post-training 8-bit quantization (INT8) for embedded Jetson/Raspberry Pi inference, and fine-tuning transfer backbones at native 224×224 resolution.

9. Conclusion

The RoadSense project demonstrates that **clean benchmark accuracy is a misleading proxy for real-world perception robustness**. While all four evaluated architectures achieved between 93.5% and 98.7% clean test accuracy, their resilience under environmental corruptions diverged dramatically. The custom 6-layer **Improved CNN (823k parameters) trained with label-safe augmentations emerged as the definitive winner**, achieving 98.70% clean accuracy and 83.11% mCA, decisively outperforming large ImageNet-pretrained transfer models. Transfer models proved disproportionately fragile under high-frequency noise and low-resolution inputs. Furthermore, native Grad-CAM diagnostics visually validated that corruption causes attention maps to collapse away from sign symbols. These findings confirm that compact, domain-specialized CNNs coupled with physics-aligned data augmentations offer superior reliability for embedded automotive vision.

10. Team Contribution, AI Declaration & References

Team Contribution Matrix:

Member	Role	Phase	Primary Ownership & Deliverables	Key Viva Focus

Aseem (Team Leader)	Leader	Phase 6	Dataset acquisition (src/download_data.py), project architecture, documentation, final technical report, repository README	End-to-end pipeline, dataset hygiene, report
Raju	Member	Phases 1–3	EDA & leakage-free track split (src/eda_and_split.py), scratch BaselineCNN, training loop, corruption test bench (src/corruptions.py, src/robustness_bench.py)	Track-based split, baseline design, 15-condition bench
Manoj	Member	Phase 4	TransferModel class (ResNet-18, MobileNetV2), two-stage fine-tuning, unified training pipeline (src/train.py rewrite), label-safe augmentation, CPU fast runner (run_fast.py)	Two-stage fine-tuning, label-safe augmentations
Sohail	Member	Phases 4 & 5	Model training execution, performance benchmarking, PyTorch native Grad-CAM engine (src/explainability.py), cross-model attention comparisons, Indian sign transfer check	Grad-CAM mechanics, attention shift, Indian signs

Declaration of Assistive Tool & AI Usage: In compliance with Atria University academic sprint guidelines, the authors declare that an AI assistant (Claude, Anthropic) was utilized as an assistive programming and typesetting tool. Specifically, the assistant was employed for code refactoring suggestions and PDF stylesheet formatting. All core research hypotheses, network architecture designs, experimental benchmarking pipelines, metric verifications, error diagnoses, and technical analyses were independently conducted, interpreted, and verified by the student authors.

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